THE CONORMAL SEQUENCE OF A PAIR OF FORMALLY UNRAMIFIED MORPHISMS IS EXACT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1383-1403, including the two xymatrix displays in the statement; proof span is lines 1405-1435.

Editorial note. The proof contains two xymatrix displays; the retained preamble provides the Xy-pic setup.

Editorial note. The cited results (the universality sequence, the two-immersions sequence, formally unramified differentials, the triangle of differentials, and the universally-unramified differentials lemma) are prerequisites and are supplied as non-load-bearing navigation stubs.

Editorial note. The original sentence 'By Lemma \ref{lemma-universally-unramified-differentials-sequence} here is a canonical morphism' drops a word before 'here'; it is retained verbatim and not repaired.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally unramified morphisms, universal first order thickenings, conormal sheaves and the conormal sequence of two immersions, and the triangle of differentials. The cited inputs are prerequisites; the construction of Z' -> Z", its placement in the commutative diagram, and the three identifications are retained.

Source reference. Original lemma lemma-differentials-universally-unramified: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1333–1342. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-formally-unramified-differentials: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 898–901. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-universal-thickening-functorial: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1220–1242. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-universal-thickening-unramified: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1179–1183. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-universally-unramified-differentials-sequence: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1360–1368. Referenced by the retained statement or proof but not part of the collected spans;

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supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-triangle-differentials: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1435–1435. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-two-immersions: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1435–1435. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let*

$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ & \searrow^{j} & \downarrow \\ & & Y \end{array}$$

be a commutative diagram of schemes where i and j are formally unramified. Then there is a canonical exact sequence

$$\mathcal{C}_{Z/Y} \to \mathcal{C}_{Z/X} \to i^*\Omega_{X/Y} \to 0$$

where the first arrow comes from Lemma lemma-universal-thickening-functorial and the second from Lemma lemma-universally-unramified-differentials-sequence.

Proof. Denote $Z \to Z'$ the universal first order thickening of Z over X. Denote $Z \to Z''$ the universal first order thickening of Z over Y. By Lemma lemma-universally-unramified-differentials-sequence here is a canonical morphism $Z' \to Z''$ so that we have a commutative diagram

$$\begin{array}{ccccc} Z & \xrightarrow{i'} & Z' & \longrightarrow & X \\ & \searrow^{j'} & \downarrow & & \downarrow \\ & & Z'' & \longrightarrow & Y \end{array}$$

Apply Morphisms, Lemma morphisms-lemma-two-immersions to the left triangle to get an exact sequence

$$\mathcal{C}_{Z/Z''} \to \mathcal{C}_{Z/Z'} \to (i')^*\Omega_{Z'/Z''} \to 0$$

As Z'' is formally unramified over Y (see Lemma lemma-universal-thickening-unramified) we have $\Omega_{Z'/Z''} = \Omega_{Z/Y}$ (by combining Lemma lemma-formally-unramified-differentials and Morphisms, Lemma morphisms-lemma-triangle-differentials). Then we have $(i')^*\Omega_{Z'/Y} = i^*\Omega_{X/Y}$ by Lemma lemma-differentials-universally-unramified. □